

Repair Bias in Data

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sampling and Representation

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Noise and Outliers

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Combine counts for Selection bias and coverage. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Selection bias and coverage. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit independently

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

6. Solve the given case

Training has eighteen adventure readers and two poetry readers; many intended users prefer poetry.

Propose repair and evaluation.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A dataset has 18 adventure readers and 2 poetry readers. A recommender ignores poetry preferences. What would you inspect and test after a repair?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 20 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Change feature coverage or weights only after locating which reader profiles are underserved.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A catalogue has 30 adventure titles and 2 poetry titles. A recommender rarely suggests poetry. Design a repair and an evaluation that distinguishes catalogue coverage from scoring preference.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Locate undercoverage

A morning-only survey claims to represent all library visitors. Which visitors may be missed?

My reasoning and evidence

13. W2 Combine counts

Group A has 3 Yes of 4 and B has 2 Yes of 6. Find the combined proportion.

My reasoning and evidence

14. W3 Repair the collection

A form is shared only in the robotics club. How can the process better cover the grade?

My reasoning and evidence

15. W4 Audit independently

A survey has 18 Yes of 20 club members and 2 Yes of 10 others. Compute both group rates and combined rate. Explain why 20/30 does not automatically describe the whole grade.

My reasoning and evidence

16. W5 Avoid causal claims

Club members favor coding more often. Does club membership necessarily cause that preference?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Repair Bias in Data

A fictional school polls learners about adding comic recommendations.

Library-club respondents: 12 Yes out of 15. Other respondents: 3 Yes out of 15.

The full grade has 30 club members and 90 others. These respondents were volunteers, not random selections.

Volunteer group comparison

	Yes responses	Respondents	Grade members
Club	12	15	30
Others	3	15	90

Read a cell using both its row and column labels.

Repair Bias in Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group rates

Club: $12/15=80\%$; others: $3/15=20\%$.



2. Respondent total

15 Yes out of 30 respondents = 50% .



3. Check representation

Respondents split 50/50, but the grade is 25% club and 75% other.



4. Repair collection

Plan including equality, documented invitations and report nonresponse by group.

CHANGE ONE THING

Recruit more club volunteers only.

Explain the changed result using the matching lesson.

Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Broaden recommender feedback collection and show respondent group coverage before making user-preference claims.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional school polls learners about adding comic recommendations.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Plan including equality, documented invitations and report nonresponse by group.

1. Group rates
2. Respondent total
3. Repair collection
4. Check representation

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Repair collection Plan including equality, documented invitations and report nonresponse by group. The combined respondent result gives club members more influence than their share of the grade. A weighted illustration using 25% and 75% gives $0.25 \times 80\% + 0.75 \times 20\% = 35\%$, but volunteers may differ within each group, so weighting alone does not establish the true grade preference. Improve the collection process rather than announcing the weighted number as fact.

Practice 2: explain the deciding evidence

Changed-case practice: Recruit more club volunteers only. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Sample size increases without fixing missing representation of other learners.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Sample size increases without fixing missing representation of other learners. The added respondents come from the already overrepresented group.

Repair a missing building block

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can 80% response be reported as 80% of the whole grade responding?. Return to this week and explain what you changed.

Review Noise and Outliers (ai-2/17). Goal: Calculate mean and median. Short practice: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?. Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf predictions from training examples. Short practice: Why does changing colour alone not change this tree's prediction?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
